one kind of complexity for another, and indeed nonlinear equations are notoriously difficult to understand fully from a theoretical point of view. However, Newton's equations involve differentiating solutions only in time, and when we discretize the equations to put them on a computer we only have to introduce a grid in this one direction. The fact that there are many equations is not so serious, as the equations themselves are of a type that can easily be treated numerically. The total number of equations we need to solve only goes up linearly with N and the work involved in solving them numerically increases in a relatively modest "polynomial" way (i.e. as a polynomial function of N), compared to the exponential dependence of the computational work on the number of particles that is typically observed in the treatment of quantum mechanics. Even though there are simplified formulations of quantum mechanics and clever discretizations that make the complexity more manageable, the price is still much higher than that of the classical alternative.

The use of an empirical energy function limits the applicability of molecular dynamics, by ruling out certain behaviors that rely explicitly on chemical reactions; even when the assumption is appropriate, the derivation of an effective classical potential energy is nontrivial and may introduce modelling errors. The function U in (1.2) may thus be exceedingly complex in its structure, and the theoretical properties of the dynamical system are not easy to explain, yet, using Newton's equations, molecular simulations involving thousands or even millions of atoms become tractable. In this book we will work with the classical model since it has widespread significance and provides a very good starting point for numerical simulation. Most of the fundamental issues surrounding the theory of molecular dynamics and the construction of numerical methods are actually independent of the detailed features of the potential energy function—there is thus a remarkable generality to the molecular level models and methods used in chemistry, physics, engineering and biology.

Using a classical description based on a smooth, empirical force field is only one of several options available for molecular simulation. The simplest model for a molecular interaction potential is the hard-sphere model. We assume that each atom is an impenetrable sphere which interacts with other atoms via perfectly elastic collision with the atoms transferring, according to standard rules, momentum and energy to one another during the collisions. This was the first model treated in simulation [4] for understanding the statistical mechanics of liquids. Hard sphere simulations are sometimes referred to as "event-driven" since the steps in time are usually taken between the collision events (for recent examples, see [107]). We can think of hard sphere systems as being described by nonsmooth potentials ("infinite at short range"), and it is possible to model systems that include both hard core repulsion and smooth attractive potentials (we describe such methods in Sect. 3.7). Relying exclusively on hard spheres is very restrictive in terms of model accuracy and leads to less efficient propagation algorithms. Thus most modern simulations are based on smooth potentials.